

Riyaz Mohammad

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Summary

Computer Science graduate with hands-on experience building and deploying AI-enabled and machine learning web applications. Skilled in designing ML inference pipelines, feature engineering workflows, model evaluation, REST API development, and full-stack software solutions using Python, Flask, React, Node.js, and SQL databases. Experienced across the Software Development Life Cycle (SDLC), including system design, development, testing, deployment, and performance optimization. Strong foundation in software engineering, object-oriented programming, distributed systems, and AI-assisted development workflows. Passionate about building scalable, user-focused, and data-driven applications.

Skills

- **Programming Languages:** Java, Python, C++, JavaScript (ES6+), SQL, HTML, CSS
- **Frontend:** HTML5, CSS3, React, Bootstrap
- **Backend:** Node.js, Spring Boot, Express.js, Flask, Spring MVC, REST APIs, Microservices, Unit & Integration Testing
- **Databases:** MySQL, PostgreSQL, MongoDB, NoSQL
- **Tools & Practices:** Git, Agile/Scrum, SDLC, CI/CD concepts, Debugging, Kubernetes, RabbitMQ, Docker, AI-Powered Application Development, AI-Assisted Development Workflows (Claude)
- **Concepts:** OOP, Design Patterns, Network Protocols, Event-Driven Architecture, Code Reviews, Performance Optimization,

Experience

BeFit – AI-Powered Fitness Application | [\[GitHub\]](#) [\[APP\]](#) **React, Node.js, Express, Python, Claude AI, REST APIs**

- Developed a full-stack fitness web application with React frontend and Node.js/Express backend supporting authentication and user data management
- Built and tested RESTful APIs handling 200+ daily requests with sub-250ms response times
- Implemented secure authentication using JWT and bcrypt for encrypted user signup/login workflows
- Designed backend data layer using structured JSON (demo mode) simulating database operations for Render deployment
- Integrated frontend and backend into a cohesive full-stack system with clean API-driven architecture

CardioCheck – Heart Risk Prediction System | [\[GitHub\]](#) [\[APP\]](#) **Python, Scikit-learn, Flask, JavaScript, HTML, CSS**

- Built an end-to-end full-stack machine learning web application with Flask backend and interactive frontend UI
- Developed REST APIs for real-time ML inference and frontend integration
- Implemented ML pipeline including preprocessing, feature engineering, and training on 68,000+ records
- Achieved 0.803 ROC-AUC using Gradient Boosting with 5-fold cross-validation
- Delivered real-time predictions (<300ms latency) via API-connected frontend interface

Classroom Management System | [\[GitHub\]](#) **Java, Spring Boot, Maven, MSSQL, HTML5/CSS**

- Architected scalable RESTful APIs using Spring Boot, reducing system latency for administrative operations by 30%.
- Optimized complex SQL queries and indexing, improving report generation speed by 25% while strengthening data security with role-based access control.
- Coordinated development using Agile practices and Git version control, ensuring efficient collaboration and timely delivery of system features.

Microservices Video Processing System | [\[GitHub\]](#) **Python, FastAPI, Docker, Kubernetes, RabbitMQ, MongoDB**

- Designed and deployed a containerized microservices system using FastAPI and Docker, orchestrated with Kubernetes for scalable, fault-tolerant video processing
- Implemented asynchronous task pipelines using RabbitMQ, enabling parallel processing and improving system throughput
- Developed modular APIs with Git-based workflows and containerized build processes, improving build reproducibility and aligning with CI/CD practices

Education

Honours Bachelor of Science in Computer Science

University: Lakehead University | **Graduated:** December 2025

Relevant Coursework: Data Structures, Algorithms, Database Management Systems, Big Data, Cloud Computing, Operating Systems, Software Engineering, Web Health Informatics, Computer Networking